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Koike et al.

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(54) **1-ARYLCARBONYL-4-OXY-PIPERIDINE COMPOUNDS USEFUL FOR THE TREATMENT OF NEURODEGENERATIVE DISEASES**

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(73) Assignee: **TAKEDA PHARMACEUTICAL COMPANY LIMITED**, Osaka (JP)

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Oct. 7, 2011 (JP) 2011/222741

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C07D 401/10 (2006.01)
C07D 401/14 (2006.01)
C07D 405/10 (2006.01)
C07D 405/14 (2006.01)
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C07D 413/14 (2006.01)
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CPC **C07D 498/08** (2013.01); **C07D 401/10** (2013.01); **C07D 401/14** (2013.01); **C07D 405/10** (2013.01); **C07D 405/14** (2013.01); **C07D 413/10** (2013.01); **C07D 413/14** (2013.01); **C07D 417/14** (2013.01)

(58) **Field of Classification Search**

CPC .. C07D 498/08; C07D 401/10; C07D 401/14; C07D 405/10; C07D 405/14; C07D 413/10; C07D 413/14; C07D 417/14

See application file for complete search history.

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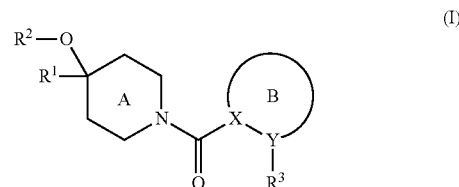
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(57) **ABSTRACT**

The present invention provides a compound useful as an agent for the prophylaxis or treatment of neurodegenerative disease and the like, or a salt thereof.

The present invention relates to a compound represented by the formula



wherein each symbol is as defined in the specification, or a salt thereof.

16 Claims, No Drawings

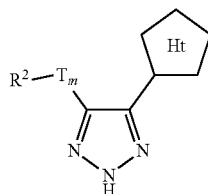
Specification includes a Sequence Listing.

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R⁶ is H, hydroxy, aryl or the like;
 X, Y and Z are independently O, NR⁷ or CR⁷₂;
 R⁷ is, H, C₁₋₈ alkyl, C₂₋₈ alkenyl, C₁₋₄ alkoxy, heteroaryl-
 C₁₋₆ alkyl, aryl-C₁₋₆ alkyl or the like; and
 n is 0-3,

as an agent for the treatment of diseases associated with
 immune disease, dementia, hypertension, diabetes and the
 like (e.g., Alzheimer's disease etc.)

Patent document 4 discloses the following compound:



wherein

Ht is a heterocyclic group (pyrrol-3-yl, [1,2,4]triazol-3-yl,
 [1,2,3]triazol-4-yl or tetrazol-5-yl, the pyrrol-3-yl has R³ and
 Qn-R⁴, and the [1,2,4]triazol-3-yl or [1,2,3]triazol-4-yl has
 R³ or Qn-R⁴);

T and Q are independently —C(O)— or the like;

m and n are independently 0-1;

R² is R or the like;

R³ is R⁷, halogen, cyano or the like;

R is a C₁₋₆ hydrocarbon group, C₆₋₁₀ aryl, C₆₋₁₀ heteroaryl,
 C₃₋₁₀ heterocycloalkyl or the like; and

R⁷ is H, an optionally substituted C₁₋₆ hydrocarbon group or
 the like,

as an agent for the treatment of autoimmune diseases,
 Alzheimer's disease and the like.

DOCUMENT LIST

Patent Document

Patent Document 1: WO 2009/117421

Patent Document 2: WO 2008/134547

Patent Document 3: WO 2008/011453

Patent Document 4: WO 02/088097

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Non-Patent Document 1: Science, vol. 261, 921-923, 1993

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SUMMARY OF THE INVENTION

Problems to be Solved by the Invention

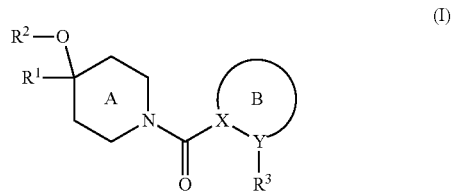
An object of the present invention is to provide a com-
 pound having a superior CH24H inhibitory action, which is
 useful as an agent for the prophylaxis or treatment of
 neurodegenerative disease (e.g., Alzheimer's disease, mild
 cognitive impairment, Huntington's disease, Parkinson's
 disease, amyotrophic lateral sclerosis, traumatic brain injury,
 cerebral infarction, glaucoma, multiple sclerosis and the
 like), epilepsy, schizophrenia and the like.

Means of Solving the Problems

The present inventors have conducted intensive studies in
 an attempt to solve the above-mentioned problem and found
 that a compound represented by the following formula (I)
 has a superior CH24H inhibitory action, which resulted in
 the completion of the present invention.

Accordingly, the present invention provides the follow-
 ing.

[1] A compound represented by the formula (I):



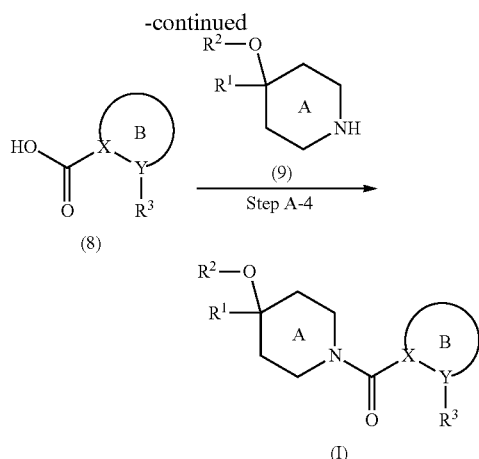
wherein

R¹ is an optionally substituted C₁₋₆ alkyl group;

R² is a hydrogen atom or an optionally substituted C₁₋₆ alkyl
 group;

R³ is an optionally substituted 5- or 6-membered aromatic
 heterocyclic group;

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wherein each symbol is as defined above.

The compound of the present invention can be produced by a sequence of reaction steps of Step A-1 to Step A-4. (Step A-1)

Compound (5) can be produced by reacting compound (2) with compound (3) or compound (4) (R³=4-pyrimidinyl). The reaction is carried out in the presence of a metal catalyst. The metal catalyst is preferably a palladium compound [e.g., palladium(II) acetate, tetrakis(triphenylphosphine)palladium(0), dichlorobis(triphenylphosphine)palladium(II), dichlorobis(triethylphosphine)palladium(II), tris(dibenzylideneacetone)dipalladium(0), a complex of palladium(II) acetate and 1,1'-bis(diphenylphosphino)ferrocene, etc.]. The amount of the metal catalyst to be used is about 0.000001-1.0 mol per 1 mol of compound (2). The metal catalyst can be used together with a phosphine ligand. The amount of the phosphine ligand to be used is about 0.01-5 mol per 1 mol of compound (2). Examples of the phosphine ligand include triphenylphosphine, 4,5-bis(diphenylphosphino)-9,9-dimethylxanthene, tri-tert-butylphosphine and the like. In addition, a salt such as tri-tert-butylphosphine tetrafluoroborate can be used. The reaction is generally carried out in the presence of a base. Examples of the base include inorganic bases, basic salts and the like. When desired, the reaction may be carried out by adding an additive such as copper(I) cyanide, copper(I) bromide and the like. The amount of compound (3) or compound (4) to be used is about 0.8-10 mol per 1 mol of compound (2). The amount of the base to be used is about 1-20 mol per 1 mol of compound (2). The amount of the additive to be used is about 0.000001-5.0 mol per 1 mol of compound (2). When a metal catalyst unstable to oxygen is used for the reaction, the reaction is preferably carried out in a stream of an inactive gas such as argon gas, nitrogen gas and the like. This reaction is advantageously carried out in a solvent inert to the reaction. While the solvent is not particularly limited as long as the reaction proceeds, preferable examples thereof include alcohols, ethers, aromatic hydrocarbons, saturated hydrocarbons, amides, halogenated hydrocarbons, nitriles, esters, water, mixed solvents thereof and the like. While the reaction time varies depending on the reagent or solvent to be used, it is generally 1 min-200 hr. The reaction temperature is preferably 0-150° C. In addition, the reaction can be carried out with irradiation of microwave in order to promote the reaction.

(Step A-2) Compound (5) can also be produced by reacting compound (6) with compound (7). The reaction is carried out in the same manner as in Step A-1.

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When desired, compound (5) produced in Step A-1 or Step A-2 can be subjected to a reduction step. For example, when compound (5) contains N-oxido or a halogen atom, it is removed by a reduction reaction known per se using palladium carbon and the like.

(Step A-3)

Compound (8) can be produced by removing the protecting group P of compound (5). The removal of the protecting group can be carried out according to a method known per se, for example, the method described in "Protective Groups in Organic Synthesis, 3rd Ed", John Wiley and Sons, Inc. (1999) (Theodora W. Greene, Peter G. M. Wuts), or the like.

Compound (8) can also be produced according to a method known per se, or a method analogous thereto.

(Step A-4)

Compound (I) can be produced by reacting carboxylic acid (8) or a reactive derivative thereof with compound (9). Examples of the reactive derivative of the carboxylic acid include acid halides such as acid chlorides, acid bromides and the like; acid amides with pyrazole, imidazole, benzotriazole and the like; mixed acid anhydrides with acetic acid, propionic acid, butyric acid and the like; acid azides; activated esters such as diethoxyphosphate ester, diphenoxyphosphate ester, p-nitrophenyl ester, 2,4-dinitrophenyl ester, cyanomethyl ester, pentachlorophenyl ester, ester with N-hydroxysuccinimide, ester with N-hydroxyphthalimide, ester with 1-hydroxybenzotriazole, ester with 6-chloro-1-hydroxybenzotriazole, ester with 1-hydroxy-1H-2-pyridone, and the like; activated thioesters such as 2-pyridyl thioester, 2-benzothiazolyl thioester and the like, and the like. Compound (I) can also be produced by directly reacting carboxylic acid (8) with compound (9) in the presence of a suitable condensing agent, instead of using the reactive derivative. Examples of the condensing agent include N,N'-disubstituted carbodiimides such as N,N'-dicyclohexylcarbodiimide,

1-ethyl-3-(3-dimethylaminopropyl)carbodiimide (WSC) hydrochloride and the like; azolides such as N,N'-carbonyldiimidazole and the like; dehydrating agents such as N-ethoxycarbonyl-2-ethoxy-1,2-dihydroquinoline, phosphorus oxychloride, alkoxyacetylene and the like; 2-halogenopyridiniums such as 2-chloromethylpyridinium iodide, 2-fluoro-1-methylpyridinium iodide and the like; phosphorilcyanides such as diethylphosphorylcyanide and the like; 2-(7-azabenzotriazol-1-yl)-1,1,3,3-tetramethyluronium hexafluorophosphate (HATU), 0-(7-azabenzotriazol-1-yl)-N,N,N',N'-tetramethyluronium tetrafluoroborate (TATU) and the like. When a condensing agent is used, the reaction is considered to progress via a reactive derivative of carboxylic acid (8). The amount of carboxylic acid (8) or a reactive derivative thereof to be used is generally about 0.8-5 mol per 1 mol of compound (9). This reaction is advantageously carried out in a solvent inert to the reaction. While the solvent is not particularly limited as long as the reaction proceeds, preferable examples thereof include ethers, aromatic hydrocarbons, saturated hydrocarbons, amides, halogenated hydrocarbons, nitriles, sulfoxides, aromatic organic bases, mixed solvents thereof. The reaction can be carried out in the presence of a basic salt, an organic bases or the like in order to promote the reaction. In addition, when an acidic substance is released due the reaction, a basic salt, an organic base and the like can be used in order to remove it from the reaction system. While the reaction time varies depending on the reagent or solvent to be used, it is generally 10 min-72 hr. The reaction temperature is preferably 0-100° C.

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Example 105

(4-(2,4-difluorobenzyl)-4-hydroxypiperidin-1-yl)
(2-(pyrimidin-4-yl)pyridin-3-yl)methanone

A suspension of 2-(pyrimidin-4-yl)nicotinic acid dihydrochloride (0.36 g), 4-(2,4-difluorobenzyl)-4-hydroxypiperidine hydrochloride (0.42 g), HATU (0.75 g) and triethylamine (1.1 mL) in DMF (5 mL) was stirred overnight at room temperature. The reaction mixture was diluted with water, and the mixture was extracted with ethyl acetate. The extract was washed with saturated brine, and dried over anhydrous sodium sulfate. The solvent was evaporated

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under reduced pressure. The residue was purified by silica gel column chromatography (NH, ethyl acetate/hexane), and recrystallized from ethyl acetate/hexane to give the title compound (0.18 g).

¹H NMR (300 MHz, CDCl₃) δ 1.29-2.01 (5H, m), 2.80 (2H, s), 3.09-3.49 (3H, m), 4.40-4.66 (1H, m), 6.76-6.91 (2H, m), 7.10-7.24 (1H, m), 7.38-7.51 (1H, m), 7.60-7.76 (1H, m), 8.17-8.29 (1H, m), 8.70-8.78 (1H, m), 8.79-8.90 (1H, m), 8.91-9.23 (1H, m).

The compounds of the examples produced according to the above-mentioned method or a method analogous thereto are shown in the following tables. MS in the tables means those found.

TABLE 1

Example No.	IUPAC Name	Chemical Structure	MS
1	(4-benzyl-4-hydroxypiperidin-1-yl) (5-methyl-2-(pyridin-4-yl)phenyl)methanone		387.2
2	(4-benzyl-4-hydroxypiperidin-1-yl) (3-methyl-5-(pyridin-4-yl)-1,2-oxazol-4-yl)methanone		378.1
3	(4-hydroxy-4-(pyridin-2-ylmethyl)piperidin-1-yl) (5-methyl-2-(pyridin-4-yl)phenyl)methanone		388.2
4	4-((4-hydroxy-1-(5-methyl-2-(pyridin-4-yl)benzoyl)piperidin-4-yl)methyl)benzonitrile		412.3

TABLE 2-continued

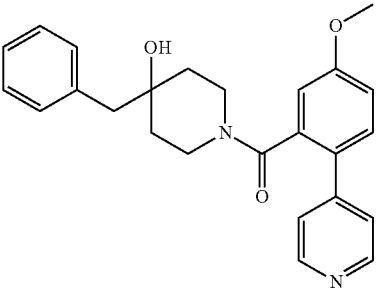
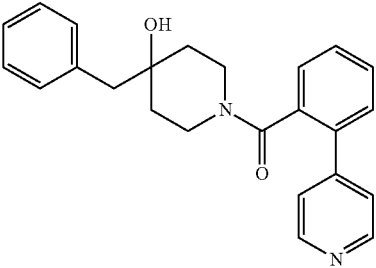
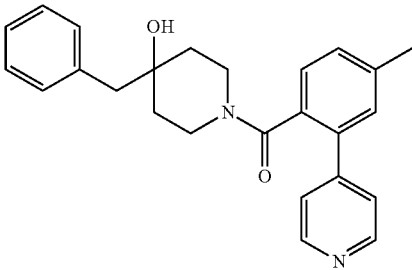
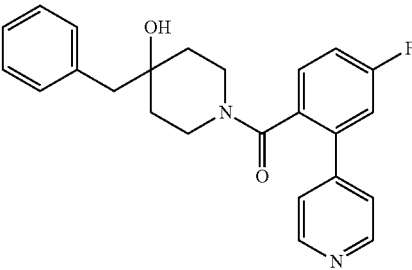
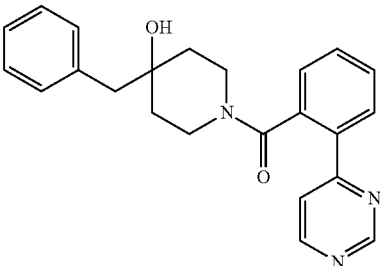
Example No.	IUPAC Name	Chemical Structure	MS
10	(4-benzyl-4-hydroxypiperidin-1-yl) (5-methoxy-2-(pyridin-4-yl)phenyl)methanone		403.2
11	(4-benzyl-4-hydroxypiperidin-1-yl) (2-(pyridin-4-yl)phenyl)methanone		373.2
12	(4-benzyl-4-hydroxypiperidin-1-yl) (4-methyl-2-(pyridin-4-yl)phenyl)methanone		387.2
13	(4-benzyl-4-hydroxypiperidin-1-yl) (4-fluoro-2-(pyridin-4-yl)phenyl)methanone		391.3
14	(4-benzyl-4-hydroxypiperidin-1-yl) (2-(pyrimidin-4-yl)phenyl)methanone		374.2

TABLE 2-continued

Example No.	IUPAC Name	Chemical Structure	MS
15	(4-benzyl-4-hydroxypiperidin-1-yl) (5-methyl-2-(pyrimidin-4-yl)phenyl)methanone		388.2
16	(4-benzyl-4-hydroxypiperidin-1-yl) (5-methyl-3-(pyridin-4-yl)-1H-pyrazol-4-yl)methanone		377.2

TABLE 3

Example No.	IUPAC Name	Chemical Structure	MS
17	(4-benzyl-4-hydroxypiperidin-1-yl) (1,5-dimethyl-3-(pyridin-4-yl)-1H-pyrazol-4-yl)methanone		391.2
18	(4-benzyl-4-hydroxypiperidin-1-yl) (3-fluoro-2-(pyridin-4-yl)phenyl)methanone		391.2
19	(4-benzyl-4-hydroxypiperidin-1-yl) (2-fluoro-6-(pyridin-4-yl)phenyl)methanone		391.2

TABLE 5-continued

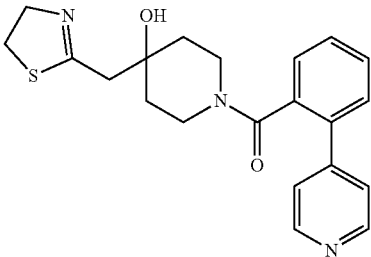
Example No.	IUPAC Name	Chemical Structure	MS
40	(4-(4,5-dihydro-1,3-thiazol-2-ylmethyl)-4-hydroxypiperidin-1-yl) (2-(pyridin-4-yl)phenyl)methanone		382.1

TABLE 6

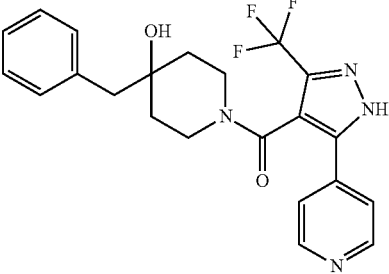
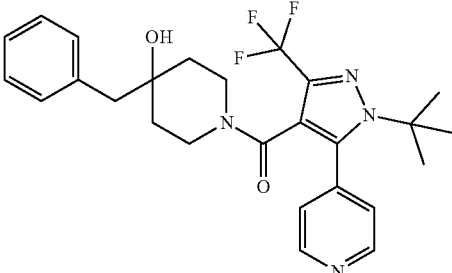
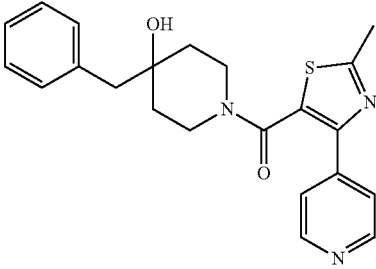
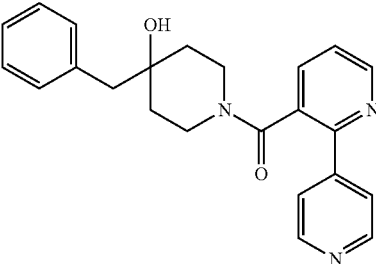
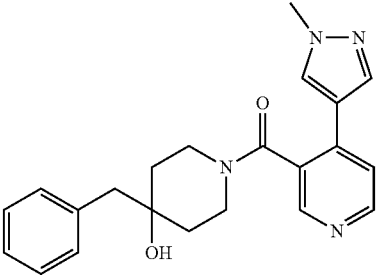
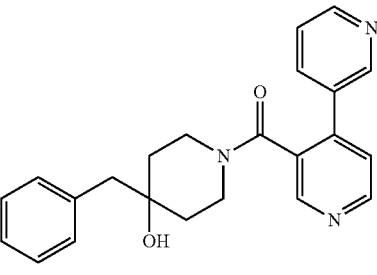
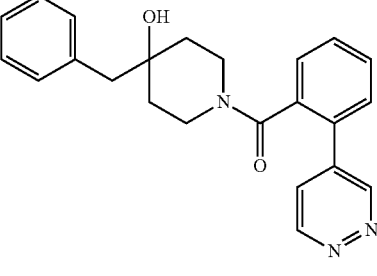
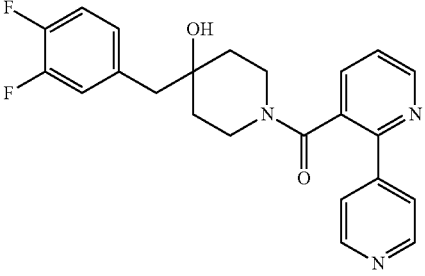
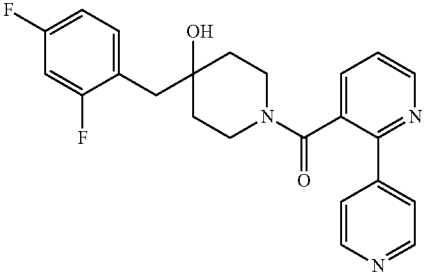
Example No.	IUPAC Name	Chemical Structure	MS
41	(4-benzyl-4-hydroxypiperidin-1-yl) (5-(pyridin-4-yl)-3-(trifluoromethyl)-1H-pyrazol-4-yl)methanone		431.1
42	(4-benzyl-4-hydroxypiperidin-1-yl) (1-tert-butyl-5-(pyridin-4-yl)-3-(trifluoromethyl)-1H-pyrazol-4-yl)methanone		487.2
43	(4-benzyl-4-hydroxypiperidin-1-yl) (2-methyl-4-(pyridin-4-yl)-1,3-thiazol-5-yl)methanone		394.2
44	(4-benzyl-4-hydroxypiperidin-1-yl) (2,4'-bipyridin-3-yl)methanone		374.2

TABLE 11-continued

Example			
No.	IUPAC Name	Chemical Structure	MS
83	(4-benzyl-4-hydroxypiperidin-1-yl) (4-(1-methyl-1H-pyrazol-4-yl)pyridin-3-yl)methanone		377.2
84	(4-benzyl-4-hydroxypiperidin-1-yl) (3,4'-bipyridin-3'-yl)methanone		374.2
85	(4-benzyl-4-hydroxypiperidin-1-yl) (2-(pyridazin-4-yl)phenyl)methanone		374.2
86	2,4'-bipyridin-3-yl (4-(3,4-difluorobenzyl)-4-hydroxypiperidin-1-yl)methanone		410.2
87	2,4'-bipyridin-3-yl (4-(2,4-difluorobenzyl)-4-hydroxypiperidin-1-yl)methanone		410.2